

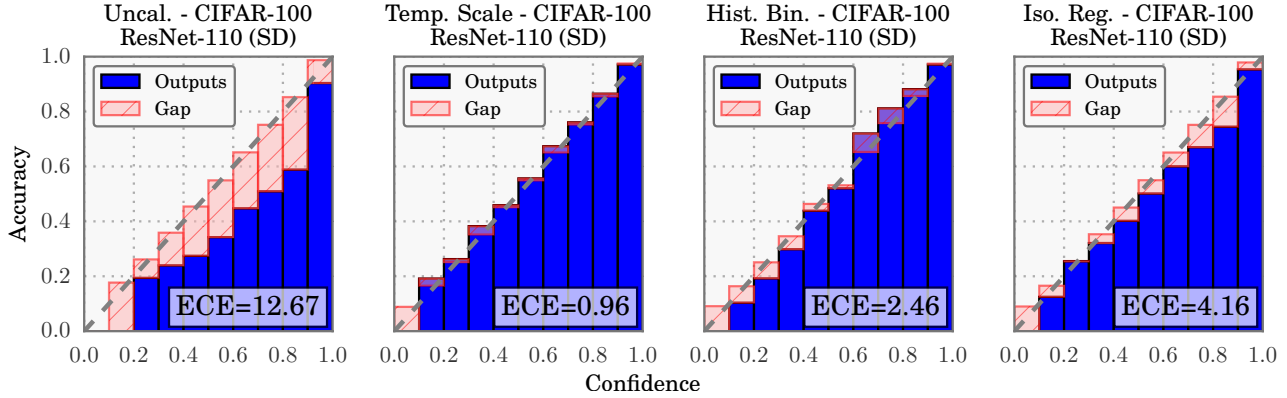


Figure 4. Reliability diagrams for CIFAR-100 before (far left) and after calibration (middle left, middle right, far right).

Matrix scaling performs poorly on datasets with hundreds of classes (i.e. Birds, Cars, and CIFAR-100), and fails to converge on the 1000-class ImageNet dataset. This is expected, since the number of parameters scales quadratically with the number of classes. Any calibration model with tens of thousands (or more) parameters will overfit to a small validation set, even when applying regularization.

Binning methods improve calibration on most datasets, but do not outperform temperature scaling. Additionally, binning methods tend to change class predictions which hurts accuracy (see Section S3). Histogram binning, the simplest binning method, typically outperforms isotonic regression and BBQ, despite the fact that both methods are strictly more general. This further supports our finding that calibration is best corrected by simple models.

Reliability diagrams. Figure 4 contains reliability diagrams for 110-layer ResNets on CIFAR-100 before and after calibration. From the far left diagram, we see that the uncalibrated ResNet tends to be overconfident in its predictions. We then can observe the effects of temperature scaling (middle left), histogram binning (middle right), and isotonic regression (far right) on calibration. All three displayed methods produce much better confidence estimates. Of the three methods, temperature scaling most closely recovers the desired diagonal function. Each of the bins are well calibrated, which is remarkable given that all the probabilities were modified by only a single parameter. We include reliability diagrams for other datasets in Section S4.

Computation time. All methods scale linearly with the number of validation set samples. Temperature scaling is by far the fastest method, as it amounts to a one-dimensional convex optimization problem. Using a conjugate gradient solver, the optimal temperature can be found in 10 iterations, or a fraction of a second on most modern hardware. In fact, even a naive line-search for the optimal temperature is faster than any of the other methods. The

computational complexity of vector and matrix scaling are linear and quadratic respectively in the number of classes, reflecting the number of parameters in each method. For CIFAR-100 ($K = 100$), finding a near-optimal vector scaling solution with conjugate gradient descent requires at least 2 orders of magnitude more time. Histogram binning and isotonic regression take an order of magnitude longer than temperature scaling, and BBQ takes roughly 3 orders of magnitude more time.

Ease of implementation. BBQ is arguably the most difficult to implement, as it requires implementing a model averaging scheme. While all other methods are relatively easy to implement, temperature scaling may arguably be the most straightforward to incorporate into a neural network pipeline. In Torch7 (Collobert et al., 2011), for example, we implement temperature scaling by inserting a `nn.MulConstant` between the logits and the softmax, whose parameter is $1/T$. We set $T=1$ during training, and subsequently find its optimal value on the validation set.⁴

6. Conclusion

Modern neural networks exhibit a strange phenomenon: probabilistic error and miscalibration worsen even as classification error is reduced. We have demonstrated that recent advances in neural network architecture and training – model capacity, normalization, and regularization – have strong effects on network calibration. It remains future work to understand why these trends affect calibration while improving accuracy. Nevertheless, simple techniques can effectively remedy the miscalibration phenomenon in neural networks. Temperature scaling is the simplest, fastest, and most straightforward of the methods, and surprisingly is often the most effective.

⁴ For an example implementation, see http://github.com/gpleiss/temperature_scaling.